

Call for Solutions – Uganda Climate Innovation Fund (UCIF) | 4th Window

Deadline – 06th October 2025

Preview

Please submit your application at this stage. See "Submit" button below.

Pioneer Details

Type

Academician / Researcher (Individual)

First Name

ASIKU

Last Name

ALLAN JUNIOR

Gender

Male

Are you a Refugee?

No

National ID

Download National ID

Country

Uganda

District(s)

Northern

GULU,

Physical Address

Kayangoga A, Bardege Layibi Division, Gulu City

Postal Address**Email Address**

arcticlinetechnologies@gmail.com

Phone Number

+256789079301

Target Geography

Lango, Acholi, West Nile, Kiryandongo, other,

Innovation Details**How did you get to know about UCIF?**

Referral by other organization or individuals,

What is your innovation?

Smart Decision Support System for Climate-Smart Agriculture and Market Insights

Areas proposed the innovation contributes to?

Access to climate smart agriculture information,

Is the proposed innovation new in Uganda?

Yes

Has the proposed innovation been fully developed and already sold in the market?

No

Describe the newness/uniqueness of the proposed innovation if it is new in Uganda and not yet available in the market.

The Smart Decision Support System (Smart DSS) introduces an AI-driven platform that integrates real-time data from weather stations, soil sensors, market trends, and pest databases to deliver personalized, actionable advice to farmers via mobile and USSD/SMS. Unlike existing platforms that provide static or generalized tips, Smart DSS uses predictive analytics and localized datasets to generate context-specific recommendations on planting, pest control, and market timing. It also integrates climate-smart practices insights, ensuring sustainable agriculture while improving productivity. The system's ability to combine multiple data sources into one adaptive intelligence model for ever-changing and evolving environmental and market dynamics makes it the first of its kind in Uganda. Its multilingual interface, offline access, and integration with mobile money for service access make it uniquely suitable for both smallholder and commercial farmers.

Why do you think it has not existed in Uganda before?

Such integrated, AI-powered decision systems require extensive data infrastructure, interoperability among agencies, and advanced analytics capabilities—resources that have only recently become available in Uganda. Most existing platforms focus on single aspects like weather or prices, lacking the holistic, adaptive intelligence Smart DSS introduces.

Climate relevance: What climate change risks and threats to the agriculture sector in Northern Uganda (NU) is the proposed innovation aiming to address or solve? How effectively does your innovation work/will work in addressing the mentioned climate change threat(s)?

Northern Uganda faces recurrent droughts, erratic rainfall, pest outbreaks, and soil degradation which are major threats to agricultural productivity and food security. These climate-related risks limit farmers' ability to plan, adapt, and sustain yields. The Smart Decision Support System (Smart DSS) directly addresses these challenges by providing real-time, data-driven insights that help farmers anticipate and respond to climatic variations. By integrating weather forecasts, soil moisture data, pest alerts, and market trends, the system offers timely recommendations on when and what to plant, irrigation scheduling, and appropriate pest management. This minimizes losses caused by unpredictable weather and enhances resource efficiency. The system through its recommendations, promotes climate-smart agricultural practices such as soil conservation and crop diversification, strengthening farmers' adaptive and anticipatory capacities. Through localized intelligence and continuous learning, Smart DSS empowers farmers to make informed decisions, improve resilience, and sustain productivity in the face of a changing climate.

Stage

Pre-prototype

Accessibility: Explain how your innovation is/will be affordable and accessible for smallholder farmers.

Smart DSS is built to be both affordable and widely accessible to smallholder farmers, especially those in rural communities. Farmers can access the system through a simple USSD code on any mobile phone, paying only a small airtime charge each time they request information—similar to standard service inquiries. This pay-per-use model makes the service flexible and affordable, eliminating the need for upfront costs or expensive devices. For farmers with smartphones, the mobile app version will offer advanced features through low-cost subscription plans, payable via mobile money. This hybrid access model ensures inclusivity across both connected and less-connected users. By leveraging existing mobile infrastructure and local telecom partnerships, Smart DSS ensures nationwide coverage without dependence on internet connectivity. The combination of minimal USSD charges, optional subscriptions, and localized language support makes the innovation practical and sustainable for Uganda's smallholder farmers.

Direct and indirect beneficiaries of the innovation

["Women Smallholder Farmers", "Refugees Smallholder Farmers", "Smallholder Farmers - General"]

Measure of success: Explain who will benefit directly and indirectly from this innovation

The innovation will directly benefit an estimated 50,000–100,000 (based various data) smallholder farmers in Northern Uganda, including women and refugee farmers, by providing timely, actionable guidance on climate-smart agricultural practices, pest management, and market opportunities. Women farmers, who often have limited access to extension services, will gain empowerment through accessible USSD and mobile app platforms in local languages, enabling them to make informed decisions and increase productivity. Refugee smallholder farmers will benefit from tailored recommendations that consider limited land, resources, and access to inputs, helping them adapt to local environmental conditions. Indirect beneficiaries include households who will experience improved food security and income stability, local cooperatives and agribusinesses that benefit from more consistent and higher-quality produce, and community support structures that gain from reduced vulnerability to climate-related losses. By enhancing adaptive capacity, resilience, and market access, Smart DSS creates broad, measurable socio-economic and environmental impacts across the agricultural ecosystem.

Other Specify

Passport

Risks

Risk One

Data gaps

Fill in with satellite data, partner with local agencies and use crowd sourced farmer data to fill missing information.

Risk Two

High adoption cost perception

Use pay-per-use USSD model and affordable mobile subscriptions using mobile money

Risk Three

Low farmer digital literacy

Conduct training via extension officers, community demos, and easy-to-follow guides

Risk Four

Risk Five

Team and partners' expertise

Name of personnel	Type of personnel	Role in the innovation	Qualification	Role in development	Declaration letter
ASIKU ALLAN JUNIOR	Innovator	Team Leader and Lead Technical Developer	BSc. Computer Science; 7 years of experience in Software Development, Systems Design, and Technology Innovation.	Leads system design, software architecture, overall technical development and innovation strategy	Not provided
ABANDU JACKSON	Innovator	Business Development and Partnerships Lead	BSc. Information Technology; 15 years of experience in ICT entrepreneurship, partnerships, and digital business strategy.	Stakeholder partnerships, Investment outreach, Strategic collaborations, market adoption	Not provided
AKELLO CAMELLA	Technical Team	IT Education and User Capacity	BSc. Information Technology Education; 10 years of	Oversees digital literacy programs, user onboarding, and trainings for	Not provided

		Development Specialist	experience in digital literacy training and ICT capacity building.	adoption and effective use	
OKOT PATRICK	Technical Team	Implementation and Support Engineer	BSc. Computer Science; 13 years of experience in mobile app development and ICT field support	Supports deployment, field integration, user assistance, system rollout and performance evaluation	Not provided

Partner's Contribution

Name of partner	Website link	Email	Phone	Role in the innovation
Assoc Prof. BENEDICT OYO	https://gepa.gu.ac.ug	b.oyo@gu.ac.ug	+256787067441	Contact Person for Integration with GEPA Smart Irrigation System Data API

Additional Documents and Videos

Additional Document(s):

[View Concept Note \(PDF\)](#)

Video:

No video uploaded.

Online Links:

No online links provided.

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Market Systems Development

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UCIF

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